

Monday
09-02-2026

EE 705: VLSI DESIGN LAB 2026
PROJECT

Note:

1. Submission is **only** through Moodle in the form of a PDF file upload.
 2. Code must be in **Verilog**
 3. **Submit all your Verilog codes and TBs.**
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Course Project: Custom Multi-Cycle RISC-V (RV32I) on FPGA

1. Objective

Design and implement a **multi-cycle pipelined RISC-V processor** that supports the **RV32I** base integer instruction set. The processor must include support for Control and Status Registers (CSRs), basic exception handling, and a memory interface. Your design must include one additional functionality as mentioned in Section-3.

2. Core Requirements (Mandatory to be implemented)

Students must adhere to the following specifications from the **RISC-V Unprivileged ISA**:

A. Instruction Set Architecture (ISA)

- **Base:** RV32I (32-bit integer base).
- **Instructions:** Must implement all R-type, I-type (including LW), S-type (SW), B-type, U-type (LUI, AUIPC), and J-type (JAL, JALR) instructions.
- **At least ONE Communication protocol (UART, SPI etc)**
- **Mandatory to implement GPIO peripheral**
- **CSR Support (Zicsr):** Implement a minimal set of Machine-mode CSRs:
 - mcycle, minstret (Performance counters)
 - mstatus, mtvec, mepc (Exception/Interrupt handling)
 - Instructions: CSRRW, CSRRS, CSRRC (Immediate variants optional).
- **Multi-cycle Multiplier (M-Extension Subset)**

B. Microarchitecture (Multi-Cycle)

- **Resource Sharing:** To fit the PYNQ-Z2 and demonstrate multi-cycle benefits, you must use **only one ALU** for both address calculation and arithmetic.
- **Memory Interface:** Implement a single memory port (Von Neumann style) that fetches instructions and data in different clock cycles.
- **State Machine:** Use a Finite State Machine (FSM) to orchestrate the *Fetch, Decode, Execute, Memory, and Write-back* stages.

C. FPGA Deployment (PYNQ Z2)

- To be deployed on PYNQ-Z2
 - **Memory:** Use on-chip **Block RAM (BRAM)** for the RISC-V program and data memory
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3. Flexibility & "Specialized Hardware" Extensions Example

1. **Custom Cryptographic Accelerator**
2. **Fixed-Point Unit**

REFERENCE:

1) [docs.riscv.org/reference/isa/ attachments/riscv-unprivileged.pdf](https://docs.riscv.org/reference/isa/attachments/riscv-unprivileged.pdf)

2) Book: Digital design and Computer architecture RISC V Edition – Sarah Harris & David Harris

Deadlines:

13 th Feb	Proposal Topic Submission deadline (Short topic)
6 th March	Detailed proposal submission. Detailed architecture, instruction supported, block diagram. Rough Timeline with deliverables. Plan on how to show the demo that the uP is working

P.S.: Initial testing has to be done using TBs on post implementation netlist. For FPGA use BRAMs and ILA to show the output. For ASIC, you can use the same testbench after making appropriate changes to your original code for Post-synthesis/Post-pnr verification.
